Admin (Snapshot 1) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Mouse integration ...

```c
#include<stdio.h>
int main()
{
        int n,r,i,j,k;
        printf("Enter number of processes: ");
        fflush(stdout);
        scanf("%d", &n);

        printf("Enter number of resource types: ");
        fflush(stdout);
        scanf("%d", &r);
        int alloc[n][r],max[n][r],need[n][r],avail[r];
        int finish[n],safe_sequence[n];
        int count=0,safeFound;
        printf("Enter allocation matrix:\n");
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        scanf("%d",&alloc[i][j]);
                }
        }
        printf("Enter max matrix:\n");
        for(i=0;i<r;i++)
        {
                for(j=0;j<r;j++)
                {
                        scanf("%d",&max[i][j]);
                }
        }
        printf("Enter available resources:\n");
        for(i=0;i<r;i++)
        {
                scanf("%d",&avail[i]);
        }
        for(i=0;i<n;i++)
        {
                finish[i]=0;
        }
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        need[i][j]=max[i][j]-alloc[i][j];
```

-- INSERT -- 1,18 Top

Right Ctrl

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Activities

Mar 21 19:11

Type to search

```
            finish[i]=0;
        }
        for(i=0;i<n;i++)
        {
                for(j=0;j<r;j++)
                {
                        need[i][j]=max[i][j]-alloc[i][j];


                }
        }
        while(count<n)
        {
                safeFound=0;

                for(i=0;i<n;i++){


                        if(finish[i]==0)
                        {
                                int safe=1;
                                for(j=0;j<r;j++)
                                {
                                        if(need[i][j]>avail[j])
                                        {
                                                safe=0;
                                                break;
                                        }
                                }
                                if(safe)
                                        {
                                        for(k=0;k<r;k++)
                                        {
                                                avail[k]=avail[k]+alloc[i][k];
                                        }

                                safe_sequence[count]=i;
                                finish[i]=1;
                                safeFound=1;
                                count++;

                                }
                        }
                }
-- INSERT --                                                    77,3-17       69%
```

Mouse integration ...

Right Ctrl

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File Machine View Input Devices Help

Mouse integration ...

```
            safeFound=0;

            for(i=0;i<n;i++){


                    if(finish[i]==0)
                    {
                            int safe=1;
                            for(j=0;j<r;j++)
                            {
                                    if(need[i][j]>avail[j])
                                    {
                                            safe=0;
                                            break;
                                    }
                            }
                            if(safe)
                                    {
                                    for(k=0;k<r;k++)
                                    {
                                            avail[k]=avail[k]+alloc[i][k];
                                    }

                            safe_sequence[count]=i;
                            finish[i]=1;
                            safeFound=1;
                            count++;

                            }
                    }
            }
            if(!safeFound)
            {
                    printf("system is in unsafe state! may occur the deadlock\n");
                    return 0;
            }
    }
    printf("safe sequence is:");
    for(i=0;i<n;i++)
    {
            printf("p%d",safe_sequence[i]);
    }
    return 0;
}
-- INSERT --                                                                    56,2-9        94%
```

hot 1) [Running] - Oracle VM VirtualBox

ew Input Devices Help

Mouse integratio...

Auto capture keyboard

Activities Terminal Mar 21 16:59

rinu@localhost:~

```
[rinu@localhost ~]$ vi banker.c
[rinu@localhost ~]$ gcc banker.c
[rinu@localhost ~]$ ./a.out
Enter number of processes: 2
Enter number of resource types: 3
1 1 0
1 1 2
Enter aloction matrix:
1 0 0
1 1 0
1 3 1
Enter max matrix:
2 1 1
7 5 3
3 3 2
Enter available resources:
1 1 2
1 1 2
safe sequence is: p0p1

[rinu@localhost ~]$
```